

Requirements Specification Document

Software Engineering - SVNC.00.308

Project: StayEase Hotel Reservation System

Type: Hotel Reservation

Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustuhov)

Date: February 20, 2026

Version: 1.0

1. Project Overview

1.1 Purpose

This system will manage room reservations for the Grand Vista Hotel, allowing guests to search available rooms, make reservations, modify or cancel bookings, and receive confirmation notifications. The goal is to eliminate manual booking errors, improve guest satisfaction through a self-service portal, and streamline front desk operations by automating reservation workflows.

1.2 Stakeholders

- Elena Petrova, Hotel General Manager (primary stakeholder, system administrator)
- Hotel Guests (primary end users, search and book rooms)
- Front Desk Staff (manage check-ins, check-outs, handle booking issues)
- Housekeeping Manager (view room turnover schedules)

1.3 Scope

In Scope

- Guest registration and authentication
- Room search with filters (date, type, capacity, price)
- Online room reservation with date selection
- Reservation modification and cancellation
- Front desk dashboard for managing reservations
- Room availability calendar view
- Email notifications for booking confirmations and cancellations
- Check-in and check-out management
- Guest reservation history
- Basic reporting for hotel manager (occupancy rates)

Out of Scope (Future Enhancements)

- Mobile application (web-based only for V1)
- Online payment processing and invoicing
- Integration with third-party booking platforms (Booking.com, Expedia)
- Room service ordering system
- Loyalty program and discount management
- SMS notifications

2. Functional Requirements

FR-1: Display Room Catalog

Description: The system shall display all room types with photos, descriptions, amenities, capacity (max guests), and nightly rates. Each room type shall show current availability status for selected dates.

Priority: High

FR-2: Guest Registration and Login

Description: The system shall allow guests to register with email, password, full name, and phone number, and log in to access booking features. Guests shall also be able to reset their password via email.

Priority: High

FR-3: Search Available Rooms

Description: The system shall allow guests to search for available rooms by specifying check-in date, check-out date, number of guests, and optionally room type. Results shall display only rooms available for the entire selected date range.

Priority: High

FR-4: Make Reservation

Description: The system shall allow logged-in guests to reserve an available room for selected dates, capturing guest details, special requests, and estimated arrival time. The system shall update room availability in real-time upon booking.

Priority: High

FR-5: Prevent Double Booking

Description: The system shall prevent reservations for rooms that are already booked for any overlapping dates, displaying a clear 'Unavailable' status and suggesting alternative rooms of the same type.

Priority: High

FR-6: Cancel Reservation

Description: The system shall allow guests to cancel their own reservations up to 24 hours before the check-in date. Cancelled reservations shall release the room back to the available inventory immediately.

Priority: High

FR-7: Modify Reservation

Description: The system shall allow guests to modify their reservation dates or room type, subject to availability. Changes shall be reflected in real-time and a confirmation email sent.

Priority: Medium

FR-8: View Reservation History

Description: The system shall display a guest's past and upcoming reservations, including room type, dates, total price, and booking status (confirmed, cancelled, completed).

Priority: Medium

FR-9: Front Desk Dashboard

Description: The system shall provide staff with a dashboard showing today's check-ins, check-outs, current occupancy, and a list of all active reservations with search and filter capabilities.

Priority: Medium

FR-10: Check-In and Check-Out

Description: The system shall allow front desk staff to mark guests as checked-in upon arrival and checked-out upon departure, updating room status accordingly (occupied, vacant, needs cleaning).

Priority: Low

FR-11: Email Notifications

Description: The system shall send automated email notifications for: booking confirmation, booking modification, booking cancellation, and a reminder 24 hours before check-in.

Priority: Low

FR-12: Occupancy Reporting

Description: The system shall generate basic occupancy reports for the hotel manager, showing occupancy rate by room type for a selected date range, average length of stay, and total reservations.

Priority: Low

3. Non-Functional Requirements

NFR-1: Performance - Booking Response Time

Description: The system shall process room search queries and return results within 2 seconds under normal load. Reservation creation shall complete within 3 seconds, including availability verification and confirmation generation.

NFR-2: Usability - Ease of Booking

Description: A first-time guest shall be able to complete a room search and reservation within 3 minutes without prior training or assistance. The interface shall follow standard web conventions and require no more than 5 steps to complete a booking.

NFR-3: Reliability - System Availability

Description: The system shall maintain 99.5% uptime, available 24/7 for guest convenience. Planned maintenance windows shall not exceed 2 hours and must be scheduled between 2:00 AM and 5:00 AM local time.

NFR-4: Security - Data Protection

Description: Guest passwords shall be hashed using bcrypt with a minimum cost factor of 12. All data transmission shall be encrypted via HTTPS (TLS 1.2+). Personal guest data (name, email, phone) shall be accessible only to authorized staff roles.

NFR-5: Scalability - Concurrent Users

Description: The system shall handle up to 100 concurrent users (guests and staff) performing searches and bookings without performance degradation below the targets specified in NFR-1.

4. Assumptions and Constraints

4.1 Assumptions

- Guests have access to internet-connected devices (desktop, tablet, or smartphone) with a modern web browser
- Hotel management will provide room inventory data (room types, capacities, rates, photos) for initial system setup
- Front desk staff will receive training on the system before launch
- Guests have valid email addresses for registration and notifications
- The hotel operates a standard reservation model (no hourly bookings)

4.2 Constraints

- Budget: Limited to \$10,000 for development and first-year hosting
- Timeline: Must be operational within 3 months, before summer peak season
- Technology: Must be web-based, compatible with existing hotel hardware (desktop PCs at front desk, tablets for management)
- Team: 3-person development team (Kyrylo, Danylo, Daniil)
- No payment processing in V1 (payments handled offline at front desk)

5. Acceptance Criteria

- All 12 functional requirements implemented and tested
- System successfully handles 100 concurrent users searching and booking rooms without errors
- Hotel manager (Elena Petrova) completes User Acceptance Testing and approves the system
- No critical or high-severity bugs in core reservation workflow (search, book, cancel, modify)
- Performance meets NFR-1 requirements (search < 2s, booking < 3s)
- Front desk staff training completed, documentation provided

6. Sign-Off

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Reviewed By: Elena Petrova (Hotel Manager) / Nicolai Morozov (Instructor)

Approval Date: February 23, 2026

Stakeholder Signature: _____ Date: _____